Basic DC Circuits — Review Sheet

Series/parallel • voltage & current drop • power loss • bleeder current • KCL/KVL | answers on p. 2, equations on p. 3

1 Series & Parallel Resistors

1. Three resistors of 220 Ω, 330 Ω, and 470 Ω are in series. Find R_{total}.
2. Resistors of 100 Ω, 220 Ω, and 330 Ω are in parallel. Find R_{total}.
3. R_1 = 100 Ω is in series with a parallel pair, R_2 = 220 Ω and R_3 = 330 Ω. Find R_{total}.

2 Voltage & Current Drop

1. A 12 V source feeds R_1 = 1 kΩ, R_2 = 2 kΩ, R_3 = 3 kΩ in series. Find the circuit current and the voltage drop across each resistor.
2. A 9 V source is across two parallel branches, R_1 = 300 Ω and R_2 = 600 Ω. Find the voltage across each, each branch current, and the total current.
3. A 24 V source feeds R_1 = 100 Ω in series with R_2 = 300 Ω ∥ R_3 = 600 Ω. Find the total current, V_{R1}, the voltage across the parallel pair, and I_2 and I_3.

3 Power Loss

1. 5 mA flows through a 2.2 kΩ resistor. How much power does it dissipate?
2. A 48 Ω resistor is connected across 12 V. Find the power dissipated. Would a ½ W resistor survive?
3. A 12 V supply delivers 10 A to a load through wires with a total resistance of 0.15 Ω. Find the power lost in the wires and the voltage that actually reaches the load.

4 Bleeder Resistor & Current Under Load

1. A 12 V supply drives a series divider: R_1 = 2 kΩ on top, R_2 = 1 kΩ on the bottom (the bleeder). With **no load**, find the bleeder current and V_{out} across R_2.
2. Same divider, but now a load R_L = 1 kΩ is connected across R_2. Find V_{out}, the load current, the bleeder current, and the current through R_1. What happened to V_{out} compared to #1?
3. Design: 12 V supply, the load needs 6 V at 2 mA. You want the bleeder to carry 10 mA. Find R_2, R_1, and the power in the bleeder.

5 KCL & KVL

1. **KCL:** At one node, 5 mA and 3 mA flow in. 2 mA flows out through one branch and I_x flows out through another. Find I_x.
2. **KVL:** A 15 V source drives three series resistors. V_{R1} = 4 V and V_{R2} = 5 V. Find V_{R3}. If R_3 = 1.2 kΩ, what is the loop current?
3. **KVL, two sources:** A 12 V source and a 5 V source are in series but opposing, with a single 700 Ω resistor in the loop. Find the loop current and the resistor voltage.

Answer Key

Quick work shown so you can see where numbers come from.

1 Series & Parallel

1. R = 220 + 330 + 470 = **1020 Ω**
2. 1/R = 1/100 + 1/220 + 1/330 = 0.01758 S, so R = ≈ **56.9 Ω**
3. 220 ∥ 330 = (220·330)/(550) = 132 Ω; R = 100 + 132 = **232 Ω**

2 Voltage & Current Drop

1. R = 6 kΩ; I = 12/6k = **2 mA**. V_1 = **2 V**, V_2 = **4 V**, V_3 = **6 V** (sum = 12 V ✓)
2. Both branches see **9 V**. I_1 = 9/300 = **30 mA**, I_2 = 9/600 = **15 mA**, I_T = **45 mA**
3. 300 ∥ 600 = 200 Ω; R_T = 300 Ω; I_T = 24/300 = **80 mA**. V_{R1} = 0.08·100 = **8 V**; parallel pair = **16 V**. I_2 = 16/300 = **53.3 mA**, I_3 = 16/600 = **26.7 mA** (sum = 80 mA ✓)

3 Power Loss

1. P = I²R = (0.005)²(2200) = **55 mW**
2. P = V²/R = 144/48 = **3 W**. No, a ½ W resistor would burn up. Use at least 5 W (ideally 2× margin).
3. Wire loss P = I²R = 10²(0.15) = **15 W**. Wire drop = 10·0.15 = 1.5 V, so the load gets 12 − 1.5 = **10.5 V**

4 Bleeder & Current Under Load

1. I_B = 12/(2k + 1k) = **4 mA**; V_{out} = 4 mA · 1k = **4 V**
2. R_2 ∥ R_L = 500 Ω; R_T = 2.5 kΩ; I_{R1} = 12/2.5k = **4.8 mA**. V_{out} = 4.8 mA · 500 = **2.4 V**. I_L = I_B = **2.4 mA** (sum = 4.8 mA ✓). V_{out} sagged from 4 V to 2.4 V under load.
3. R_2 = 6 V/10 mA = **600 Ω**. I_{R1} = 10 + 2 = 12 mA; V_{R1} = 12 − 6 = 6 V; R_1 = 6/12 mA = **500 Ω**. $P_{bleeder}$ = 6 V · 10 mA = **60 mW**

5 KCL & KVL

1. In = out: 5 + 3 = 2 + I_x, so I_x = **6 mA**
2. 15 − 4 − 5 − V_{R3} = 0, so V_{R3} = **6 V**. I = 6/1.2k = **5 mA**
3. 12 − 5 − V_R = 0, so V_R = **7 V**; I = 7/700 = **10 mA**

Reference: Equations

Everything you need for the questions above.

Basics		
Ohm's law	$V = I \cdot R \quad I = V/R \quad R = V/I$	V volts, I amps, R ohms
Power	$P = V \cdot I = I^2R = V^2/R$	P in watts
Series & Parallel		
Series resistance	$R_T = R_1 + R_2 + R_3 + \ldots$	Same current everywhere
Parallel resistance	$1/R_T = 1/R_1 + 1/R_2 + 1/R_3 + \ldots$	Same voltage across each
Two in parallel	$R_T = (R_1 \cdot R_2) / (R_1 + R_2)$	Product over sum
N equal in parallel	$R_T = R / N$	Less than smallest R
Dividers		
Voltage divider	$V_{R2} = V_S \cdot R_2 / (R_1 + R_2)$	Unloaded
Current divider (2 branches)	$I_1 = I_T \cdot R_2 / (R_1 + R_2)$	More current through smaller R
Bleeder / Loaded Divider		
Bleeder current (no load)	$I_B = V_S / (R_1 + R_2)$	R_2 is the bleeder
Loaded output	$V_{out} = V_S \cdot (R_2 \parallel R_L) / (R_1 + R_2 \parallel R_L)$	Load pulls V_{out} down
Current in R_1	$I_{R1} = I_B + I_L$	R_1 carries both
Design from I_B	$R_2 = V_{out}/I_B$; $R_1 = (V_S - V_{out})/(I_B + I_L)$	Bigger I_B = stiffer, wastes more
Kirchhoff's Laws		
KCL	$\Sigma I_{in} = \Sigma I_{out}$ (Σ I at a node = 0)	Current in = current out
KVL	Σ V around a closed loop = 0	Sources = sum of drops
Series loop form	$V_S = V_1 + V_2 + V_3 + \ldots$	Opposing sources subtract
Power Loss Tips		
Wire / line loss	$P_{loss} = I^2 \cdot R_{wire}$; $V_{drop} = I \cdot R_{wire}$	Loss grows with I^2
Resistor rating	$P_{rating} \geq 2 \times P_{calc}$	Common safety margin
Total power	$P_T = V_S \cdot I_T = \Sigma P_{each}$	Source = sum dissipated

Prefixes: k = 10^3 m = 10^{-3} μ = 10^{-6} | 1 mA = 0.001 A | 1 kΩ = 1000 Ω